Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Fagundes Bros. Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
14478 Cox Ferry Road	Snelling	Merced	95369

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1953

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X200-X036-XXXX X042-X200-X060-XXXX

B. OPERATORS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Operator name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Legal owner name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	740	75	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	740	75	0	0	0	0
Average number	740	75	0	0	0	0
Avg live weight (lbs)	1,300	1,400	0	0		

Predominant milk cow breed: Holstein

Average milk production: 61 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,276.45 *tons per reporting period*

Total nitrogen from manure: 251,649.27 *lbs per reporting period*

After ammonia losses (30% loss applied): 176,154.49 *lbs per reporting period*

Total phosphorus from manure: 41,943.26 *lbs per reporting period*

Total potassium from manure: 137,341.21 *lbs per reporting period*

Total salt from manure: 365,675.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 19,680,000 *gallons*

Total nitrogen generated: 23,550.74 *lbs*

Total phosphorus generated: 8,199.75 *lbs*

Total potassium generated: 74,196.38 *lbs*

Total salt generated: 288,266.01 *lbs*

	19,680,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	19,680,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - 60-WD	Ground water
Irrigation Water - Surface Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
42B	42	42	0	none	X042-X200-X060-XXXX
44A	44	44	0	none	X042-X200-X060-XXXX
53A	1	1	0	none	X042-X200-X006-XXXX
53B	19	19	2	manure	X042-X200-X060-XXXX
53C	20	20	2	manure	X042-X200-X060-XXXX
54	43	43	3	process wastewater	X042-X200-X060-XXXX
55A	19	19	2	manure	X042-X200-X060-XXXX
55B	20	20	2	manure	X042-X200-X060-XXXX
56B	60	60	3	manure	X042-X200-X006-XXXX
57	57	57	3	manure	X042-X160-X045-XXXX X042-X200-X060-XXXX
59A	44	44	3	manure	X042-X200-X060-XXXX
59B	42	42	2	manure	X042-X200-X060-XXXX
60	34	34	3	manure	X042-X160-X045-XXXX
House #1	10	10	0	none	X042-X200-X036-XXXX
Richards	78	78	3	manure	X042-X220-X012-XXXX
Totals for areas that were used for application	436	436	28		
Totals for areas that were not used for application	97	97	0		
Land application area totals	533	533	28		

B. CROPS AND HARVESTS

53B

Field name: 53B

53B

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	230.00 *ton*	Dry-weight		73.4	15,000.00	3,000.00	19,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.11	96.60	19.32	126.87	399.28

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	489.00 *ton*	Dry-weight		48.1	12,500.00	3,800.00	16,300.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.74	333.94	101.52	435.45	1,923.47

53C

Field name: 53C

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	256.00 *ton*	Dry-weight		73.4	15,000.00	3,000.00	19,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.80	102.14	20.43	134.15	422.20

53C

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	524.00 *ton*	Dry-weight		48.1	12,500.00	3,800.00	16,300.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.20	339.95	103.34	443.29	1,958.08

54

Field name: 54

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	533.00 *ton*	Dry-weight		74.9	17,200.00	3,100.00	20,000.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.40	107.03	19.29	124.45	435.57

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	993.00 *ton*	Dry-weight		44.5	13,600.00	3,100.00	19,800.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.09	348.61	79.46	507.54	1,896.86

54

09/01/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 43 Plant date: 09/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	538.00 *ton*	Dry-weight		84.6	21,000.00	4,200.00	21,800.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	12.51	80.93	16.19	84.01	485.55

55A

Field name: 55A

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	240.00 *ton*	Dry-weight		76.9	16,200.00	3,300.00	20,200.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.63	94.54	19.26	117.88	426.01

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/25/2023	575.00 *ton*	Dry-weight		75.1	14,800.00	3,400.00	14,800.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.26	223.05	51.24	223.05	1,039.90

55B

Field name: 55B

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	259.00 *ton*	Dry-weight		76.9	16,200.00	3,300.00	20,200.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.95	96.92	19.74	120.85	436.75

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/25/2023	598.00 *ton*	Dry-weight		75.1	14,800.00	3,400.00	14,800.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.47	231.97	53.29	231.97	1,081.50

56B

Field name: 56B

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	743.00 *ton*	Dry-weight		73.3	16,400.00	2,600.00	19,700.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.38	108.45	17.19	130.27	502.57

56B

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/10/2023	1,605.00 *ton*	Dry-weight		66.5	10,900.00	3,700.00	15,800.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.75	195.36	66.31	283.18	1,451.72

08/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 60 Plant date: 08/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	650.00 *ton*	Dry-weight		78.1	14,000.00	4,000.00	27,600.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.83	66.43	18.98	130.96	465.01

57

Field name: 57

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	700.00 *ton*	Dry-weight		69.9	13,800.00	2,900.00	18,800.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.28	102.02	21.44	138.99	487.94

57

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	1,325.00 *ton*	Dry-weight		63.9	12,900.00	3,100.00	18,800.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.25	216.51	52.03	315.53	1,057.35

09/01/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 57 Plant date: 09/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	588.00 *ton*	Dry-weight		82.5	20,300.00	3,300.00	32,500.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.32	73.29	11.91	117.34	444.09

59A

Field name: 59A

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 44 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	553.00 *ton*	Dry-weight		70.7	20,700.00	3,200.00	24,700.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.57	152.45	23.57	181.91	589.20

59A

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/11/2023	1,328.00 *ton*	Dry-weight		67.8	14,400.00	3,100.00	11,900.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.18	279.89	60.25	231.30	1,224.54

08/23/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 44 Plant date: 08/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	501.00 *ton*	Dry-weight		85.6	27,500.00	3,100.00	34,800.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.39	90.18	10.17	114.12	354.16

59B

Field name: 59B

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 42 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	527.00 *ton*	Dry-weight		75.2	13,400.00	2,800.00	20,500.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.55	83.40	17.43	127.58	392.09

59B

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,263.00 *ton*	Dry-weight		70.9	12,100.00	3,400.00	16,400.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.07	211.77	59.51	287.03	1,260.11

60

Field name: 60

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	411.00 *ton*	Dry-weight		75.3	20,900.00	4,500.00	26,800.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.09	124.81	26.87	160.04	465.78

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/10/2023	1,027.00 *ton*	Dry-weight		67.9	9,900.00	3,100.00	14,400.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.21	191.98	60.12	279.25	1,473.81

60

08/24/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 34 Plant date: 08/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	364.00 *ton*	Dry-weight		72.5	17,500.00	2,800.00	27,500.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.71	103.04	16.49	161.93	635.93

Richards

Field name: Richards

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 78 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	917.00 *ton*	Dry-weight		68.6	11,300.00	2,400.00	16,700.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.76	83.43	17.72	123.30	442.98

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/11/2023	2,155.00 *ton*	Dry-weight		70.5	13,800.00	3,900.00	16,500.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.63	224.95	63.57	268.96	1,059.54

Richards

08/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 78 Plant date: 08/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	854.00 *ton*	Dry-weight		81.5	15,000.00	3,900.00	31,500.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.95	60.77	15.80	127.61	449.66

NUTRIENT BUDGET

A. LAND APPLICATIONS

53B - 11/17/2022: Oats, silage-soft dough

Field name: 53B

Crop: Oats, silage-soft dough — Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.20	2,029,480.00 *gal*
Application event totals		0.00	0.00	0.00	31.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.03	2,083,360.00 *gal*
Application event totals		0.00	0.00	0.00	32.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.44	2,110,300.00 *gal*
Application event totals		0.00	0.00	0.00	32.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	93.93	18.90	63.92	0.00	240.00 *ton*
Application event totals		93.93	18.90	63.92	0.00	

53B - 05/16/2023: Corn, silage

Field name: 53B

Crop: Corn, silage — Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

53B - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	364.60	66.69	140.06	0.00	550.00 *ton*
Application event totals		364.60	66.69	140.06	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.00	2,406,640.00 *gal*
Application event totals		0.00	0.00	0.00	37.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.27	2,424,600.00 *gal*
Application event totals		0.00	0.00	0.00	37.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.82	2,460,520.00 *gal*
Application event totals		0.00	0.00	0.00	37.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.65	2,514,400.00 *gal*
Application event totals		0.00	0.00	0.00	38.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.96	2,469,500.00 *gal*
Application event totals		0.00	0.00	0.00	37.96	

53B - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.38	2,496,440.00 *gal*
Application event totals		0.00	0.00	0.00	38.38	

53C - 11/17/2022: Oats, silage-soft dough

Field name: 53C

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.21	2,137,240.00 *gal*
Application event totals		0.00	0.00	0.00	31.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	30.95	2,119,280.00 *gal*
Application event totals		0.00	0.00	0.00	30.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.00	2,191,120.00 *gal*
Application event totals		0.00	0.00	0.00	32.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	111.54	22.44	75.90	0.00	300.00 *ton*
Application event totals		111.54	22.44	75.90	0.00	

53C - 05/16/2023: Corn, silage

Field name: 53C

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	358.96	65.66	137.89	0.00	570.00 *ton*
Application event totals		358.96	65.66	137.89	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	35.28	2,415,620.00 *gal*
Application event totals		0.00	0.00	0.00	35.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	34.88	2,388,680.00 *gal*
Application event totals		0.00	0.00	0.00	34.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	34.49	2,361,740.00 *gal*
Application event totals		0.00	0.00	0.00	34.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	35.67	2,442,560.00 *gal*
Application event totals		0.00	0.00	0.00	35.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	34.75	2,379,700.00 *gal*
Application event totals		0.00	0.00	0.00	34.75	

53C - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	35.28	2,415,620.00 *gal*
Application event totals		0.00	0.00	0.00	35.28	

54 - 11/17/2022: Oats, silage-soft dough

Field name: 54

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.59	19.98	285.22	707.38	2,916,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.05	4,570,820.00 *gal*
Application event totals		56.59	19.98	285.22	738.43	

02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.51	19.73	281.70	698.65	2,880,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.47	4,633,680.00 *gal*
Application event totals		56.51	19.73	281.70	730.13	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.78	4,678,580.00 *gal*
Application event totals		0.00	0.00	0.00	31.78	

54 - 05/16/2023: Corn, silage

54 - 05/16/2023: Corn, silage

Field name: 54

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.28	5,783,120.00 *gal*
Application event totals		0.00	0.00	0.00	39.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.28	5,783,120.00 *gal*
Application event totals		0.00	0.00	0.00	39.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.63	37.50	244.60	1,210.04	2,676,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.85	5,720,260.00 *gal*
Application event totals		70.63	37.50	244.60	1,248.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	144.43	33.98	227.70	1,073.97	3,024,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.12	5,612,500.00 *gal*
Application event totals		144.43	33.98	227.70	1,112.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	131.53	30.95	207.37	978.08	2,754,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.88	5,576,580.00 *gal*
Application event totals		131.53	30.95	207.37	1,015.96	

54 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.24	5,630,460.00 *gal*
Application event totals		0.00	0.00	0.00	38.24	

54 - 09/01/2023: Sudangrass, silage

Field name: 54

Crop: Sudangrass, silage

Plant date: 09/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.20	36.74	239.67	1,185.62	2,622,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.26	5,926,800.00 *gal*
Application event totals		69.20	36.74	239.67	1,225.88	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	18.80	11.83	239.23	850.12	2,808,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.81	6,007,620.00 *gal*
Application event totals		18.80	11.83	239.23	890.93	

55A - 11/16/2022: Oats, silage-soft dough

Field name: 55A

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

55A - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	29.13	1,894,780.00 *gal*
Application event totals		0.00	0.00	0.00	29.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	30.09	1,957,640.00 *gal*
Application event totals		0.00	0.00	0.00	30.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	29.27	1,903,760.00 *gal*
Application event totals		0.00	0.00	0.00	29.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	117.41	23.62	79.89	0.00	300.00 *ton*
Application event totals		117.41	23.62	79.89	0.00	

55A - 05/16/2023: Corn, silage

Field name: 55A

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	265.16	48.51	101.86	0.00	400.00 *ton*
Application event totals		265.16	48.51	101.86	0.00	

55A - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.00	2,406,640.00 *gal*
Application event totals		0.00	0.00	0.00	37.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	36.31	2,361,740.00 *gal*
Application event totals		0.00	0.00	0.00	36.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	36.17	2,352,760.00 *gal*
Application event totals		0.00	0.00	0.00	36.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	36.72	2,388,680.00 *gal*
Application event totals		0.00	0.00	0.00	36.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.00	2,406,640.00 *gal*
Application event totals		0.00	0.00	0.00	37.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	36.44	2,370,720.00 *gal*
Application event totals		0.00	0.00	0.00	36.44	

55B - 11/16/2022: Oats, silage-soft dough

55B - 11/16/2022: Oats, silage-soft dough

Field name: 55B

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	29.90	2,047,440.00 *gal*
Application event totals		0.00	0.00	0.00	29.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	28.72	1,966,620.00 *gal*
Application event totals		0.00	0.00	0.00	28.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	29.38	2,011,520.00 *gal*
Application event totals		0.00	0.00	0.00	29.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	115.26	23.19	78.43	0.00	310.00 *ton*
Application event totals		115.26	23.19	78.43	0.00	

55B - 05/16/2023: Corn, silage

Field name: 55B

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

55B - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	251.90	46.08	96.77	0.00	380.00 *ton*
Application event totals		251.90	46.08	96.77	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.82	2,460,520.00 *gal*
Application event totals		0.00	0.00	0.00	37.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.55	2,442,560.00 *gal*
Application event totals		0.00	0.00	0.00	37.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.10	2,478,480.00 *gal*
Application event totals		0.00	0.00	0.00	38.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.38	2,496,440.00 *gal*
Application event totals		0.00	0.00	0.00	38.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.79	2,523,380.00 *gal*
Application event totals		0.00	0.00	0.00	38.79	

55B - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.24	2,487,460.00 *gal*
Application event totals		0.00	0.00	0.00	38.24	

56B - 11/16/2022: Oats, silage-soft dough

Field name: 56B

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.65	6,708,060.00 *gal*
Application event totals		0.00	0.00	0.00	32.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.83	6,743,980.00 *gal*
Application event totals		0.00	0.00	0.00	32.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.52	6,681,120.00 *gal*
Application event totals		0.00	0.00	0.00	32.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	117.74	23.69	80.12	0.00	950.00 *ton*
Application event totals		117.74	23.69	80.12	0.00	

56B - 05/15/2023: Corn, silage

Field name: 56B

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	251.90	46.08	96.77	0.00	1,200.00 *ton*
Application event totals		251.90	46.08	96.77	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.32	8,692,640.00 *gal*
Application event totals		0.00	0.00	0.00	42.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.14	8,656,720.00 *gal*
Application event totals		0.00	0.00	0.00	42.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.10	8,647,740.00 *gal*
Application event totals		0.00	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.36	8,701,620.00 *gal*
Application event totals		0.00	0.00	0.00	42.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.14	8,656,720.00 *gal*
Application event totals		0.00	0.00	0.00	42.14	

56B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.01	8,629,780.00 *gal*
Application event totals		0.00	0.00	0.00	42.01	

56B - 08/25/2023: Sudangrass, silage

Field name: 56B

Crop: Sudangrass, silage

Plant date: 08/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	86.75	17.45	59.03	0.00	700.00 *ton*
Application event totals		86.75	17.45	59.03	0.00	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.62	8,755,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.62	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.36	8,701,620.00 *gal*
Application event totals		0.00	0.00	0.00	42.36	

57 - 11/17/2022: Oats, silage-soft dough

Field name: 57

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

57 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.34	6,115,380.00 *gal*
Application event totals		0.00	0.00	0.00	31.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.98	6,241,100.00 *gal*
Application event totals		0.00	0.00	0.00	31.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.21	6,286,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	110.89	22.31	75.46	0.00	850.00 *ton*
Application event totals		110.89	22.31	75.46	0.00	

57 - 05/15/2023: Corn, silage

Field name: 57

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	251.90	46.08	96.77	0.00	1,140.00 *ton*
Application event totals		251.90	46.08	96.77	0.00	

57 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.09	8,019,140.00 *gal*
Application event totals		0.00	0.00	0.00	41.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.81	7,965,260.00 *gal*
Application event totals		0.00	0.00	0.00	40.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.28	8,055,060.00 *gal*
Application event totals		0.00	0.00	0.00	41.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.86	7,974,240.00 *gal*
Application event totals		0.00	0.00	0.00	40.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.72	7,947,300.00 *gal*
Application event totals		0.00	0.00	0.00	40.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.09	8,019,140.00 *gal*
Application event totals		0.00	0.00	0.00	41.09	

57 - 09/01/2023: Sudangrass, silage

57 - 09/01/2023: Sudangrass, silage

Field name: 57

Crop: Sudangrass, silage

Plant date: 09/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	71.75	14.44	48.82	0.00	550.00 *ton*
Application event totals		71.75	14.44	48.82	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.82	7,381,560.00 *gal*
Application event totals		0.00	0.00	0.00	37.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.56	7,525,240.00 *gal*
Application event totals		0.00	0.00	0.00	38.56	

59A - 11/18/2022: Oats, silage-soft dough

Field name: 59A

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	33.14	4,992,880.00 *gal*
Application event totals		0.00	0.00	0.00	33.14	

59A - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	33.80	5,091,660.00 *gal*
Application event totals		0.00	0.00	0.00	33.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	33.44	5,037,780.00 *gal*
Application event totals		0.00	0.00	0.00	33.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	169.00	34.00	115.00	0.00	1,000.00 *ton*
Application event totals		169.00	34.00	115.00	0.00	

59A - 05/15/2023: Corn, silage

Field name: 59A

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	357.82	65.45	137.45	0.00	1,250.00 *ton*
Application event totals		357.82	65.45	137.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.15	5,747,200.00 *gal*
Application event totals		0.00	0.00	0.00	38.15	

59A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.91	5,711,280.00 *gal*
Application event totals		0.00	0.00	0.00	37.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.33	5,774,140.00 *gal*
Application event totals		0.00	0.00	0.00	38.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.43	5,639,440.00 *gal*
Application event totals		0.00	0.00	0.00	37.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.09	5,738,220.00 *gal*
Application event totals		0.00	0.00	0.00	38.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.91	5,711,280.00 *gal*
Application event totals		0.00	0.00	0.00	37.91	

59A - 08/23/2023: Sudangrass, silage

Field name: 59A

Crop: Sudangrass, silage

Plant date: 08/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

59A - 08/23/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	104.78	21.08	71.30	0.00	620.00 *ton*
Application event totals		104.78	21.08	71.30	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.24	6,061,500.00 *gal*
Application event totals		0.00	0.00	0.00	40.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.77	6,142,320.00 *gal*
Application event totals		0.00	0.00	0.00	40.77	

59B - 11/18/2022: Oats, silage-soft dough

Field name: 59B

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	28.48	4,094,880.00 *gal*
Application event totals		0.00	0.00	0.00	28.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	28.85	4,148,760.00 *gal*
Application event totals		0.00	0.00	0.00	28.85	

59B - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	30.23	4,346,320.00 *gal*
Application event totals		0.00	0.00	0.00	30.23	

10/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	93.84	18.88	63.85	0.00	530.00 *ton*
Application event totals		93.84	18.88	63.85	0.00	

59B - 06/15/2023: Corn, silage

Field name: 59B

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.53	5,540,660.00 *gal*
Application event totals		0.00	0.00	0.00	38.53	

04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	275.89	50.47	105.98	0.00	920.00 *ton*
Application event totals		275.89	50.47	105.98	0.00	

06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.09	5,477,800.00 *gal*
Application event totals		0.00	0.00	0.00	38.09	

59B - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.97	5,603,520.00 *gal*
Application event totals		0.00	0.00	0.00	38.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.59	5,549,640.00 *gal*
Application event totals		0.00	0.00	0.00	38.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Domestic Well - 60-WD	Ground water	0.00	0.00	0.00	227.43	5,513,720.00 *gal*
Application event totals		0.00	0.00	0.00	227.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.16	5,630,460.00 *gal*
Application event totals		0.00	0.00	0.00	39.16	

60 - 11/18/2022: Oats, silage-soft dough

Field name: 60

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	28.70	3,340,560.00 *gal*
Application event totals		0.00	0.00	0.00	28.70	

60 - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	29.16	3,394,440.00 *gal*
Application event totals		0.00	0.00	0.00	29.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	28.85	3,358,520.00 *gal*
Application event totals		0.00	0.00	0.00	28.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	153.09	30.80	104.18	0.00	700.00 *ton*
Application event totals		153.09	30.80	104.18	0.00	

60 - 05/15/2023: Corn, silage

Field name: 60

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	207.77	41.80	141.38	0.00	950.00 *ton*
Application event totals		207.77	41.80	141.38	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.19	4,678,580.00 *gal*
Application event totals		0.00	0.00	0.00	40.19	

60 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.19	4,795,320.00 *gal*
Application event totals		0.00	0.00	0.00	41.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.35	4,813,280.00 *gal*
Application event totals		0.00	0.00	0.00	41.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.65	4,732,460.00 *gal*
Application event totals		0.00	0.00	0.00	40.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.19	4,795,320.00 *gal*
Application event totals		0.00	0.00	0.00	41.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.04	4,777,360.00 *gal*
Application event totals		0.00	0.00	0.00	41.04	

60 - 08/24/2023: Sudangrass, silage

Field name: 60

Crop: Sudangrass, silage

Plant date: 08/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

60 - 08/24/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	131.22	26.40	89.29	0.00	600.00 *ton*
Application event totals		131.22	26.40	89.29	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.43	4,939,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.66	4,849,200.00 *gal*
Application event totals		0.00	0.00	0.00	41.66	

Richards - 11/18/2022: Oats, silage-soft dough

Field name: Richards

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	31.57	8,432,220.00 *gal*
Application event totals		0.00	0.00	0.00	31.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.82	8,764,480.00 *gal*
Application event totals		0.00	0.00	0.00	32.82	

Richards - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.35	8,638,760.00 *gal*
Application event totals		0.00	0.00	0.00	32.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	95.33	19.18	64.87	0.00	1,000.00 *ton*
Application event totals		95.33	19.18	64.87	0.00	

Richards - 05/16/2023: Corn, silage

Field name: Richards

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	242.22	44.31	93.05	0.00	1,500.00 *ton*
Application event totals		242.22	44.31	93.05	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.86	11,180,100.00 *gal*
Application event totals		0.00	0.00	0.00	41.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.30	11,296,840.00 *gal*
Application event totals		0.00	0.00	0.00	42.30	

Richards - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.76	11,153,160.00 *gal*
Application event totals		0.00	0.00	0.00	41.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.83	11,171,120.00 *gal*
Application event totals		0.00	0.00	0.00	41.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.97	11,207,040.00 *gal*
Application event totals		0.00	0.00	0.00	41.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.86	11,180,100.00 *gal*
Application event totals		0.00	0.00	0.00	41.86	

Richards - 08/25/2023: Sudangrass, silage

Field name: Richards

Crop: Sudangrass, silage

Plant date: 08/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	76.27	15.34	51.90	0.00	800.00 *ton*
Application event totals		76.27	15.34	51.90	0.00	

Richards - 08/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.54	11,359,700.00 *gal*
Application event totals		0.00	0.00	0.00	42.54	

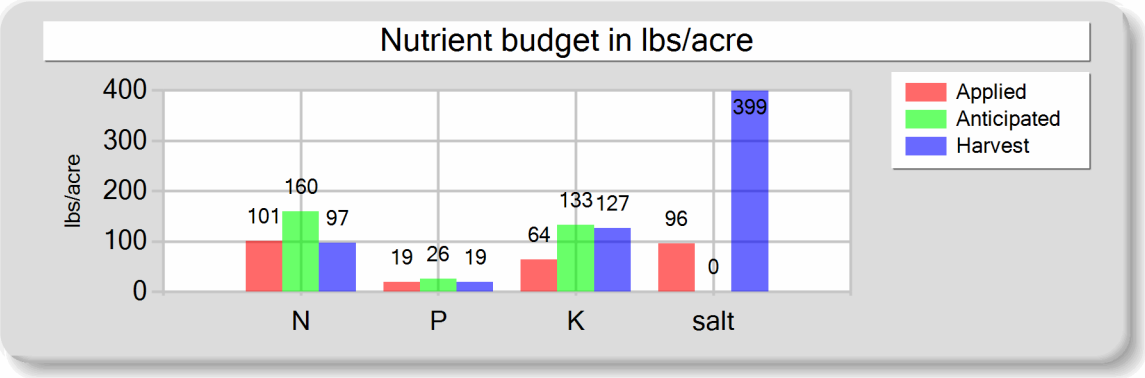
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.34	11,305,820.00 *gal*
Application event totals		0.00	0.00	0.00	42.34	

B. NUTRIENT BUDGET

53B - 11/17/2022: Oats, silage-soft dough

Field name: 53B | Crop: Oats, silage-soft dough | Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	93.93	18.90	63.92	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	95.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	100.93	18.90	63.92	95.66
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	96.60	19.32	126.87	399.28
Nutrient balance	4.33	-0.42	-62.95	-303.62
Applied to removed ratio	1.04	0.98	0.50	0.24

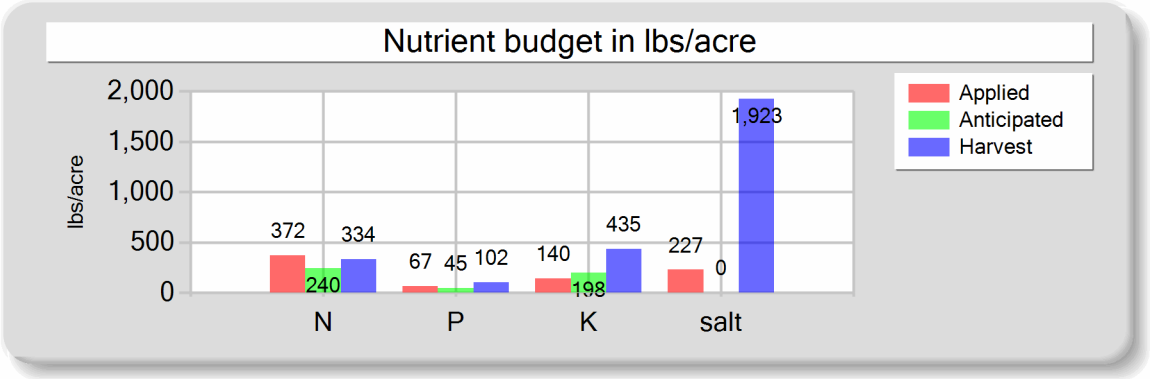
Fresh water applied	
6,223,140.00	*gallons*
229.18	*acre-inches*
12.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

53B - 05/16/2023: Corn, silage

Field name: 53B | Crop: Corn, silage | Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	364.60	66.69	140.06	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	227.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	371.60	66.69	140.06	227.08
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	333.94	101.52	435.45	1,923.47
Nutrient balance	37.66	-34.82	-295.39	-1,696.39
Applied to removed ratio	1.11	0.66	0.32	0.12

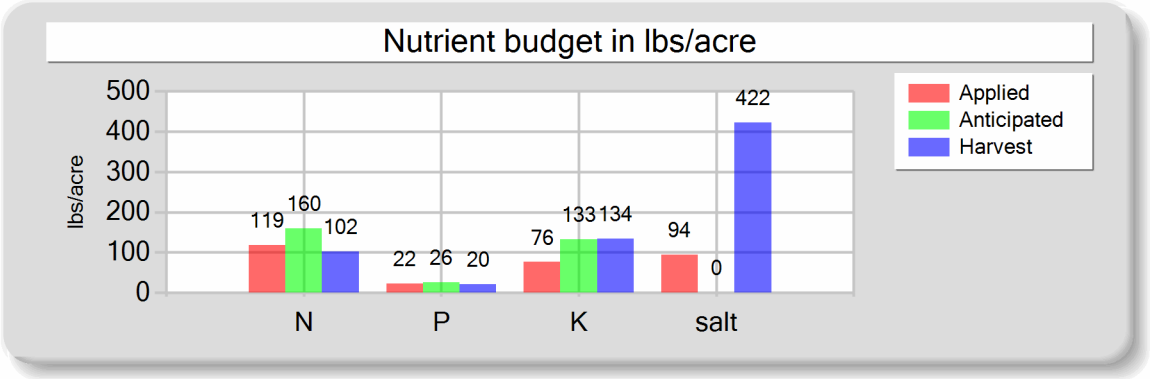
Fresh water applied	
14,772,100.00	*gallons*
544.01	*acre-inches*
28.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

53C - 11/17/2022: Oats, silage-soft dough

Field name: 53C Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.54	22.44	75.90	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	94.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.54	22.44	75.90	94.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	102.14	20.43	134.15	422.20
Nutrient balance	16.40	2.01	-58.25	-328.04
Applied to removed ratio	1.16	1.10	0.57	0.22

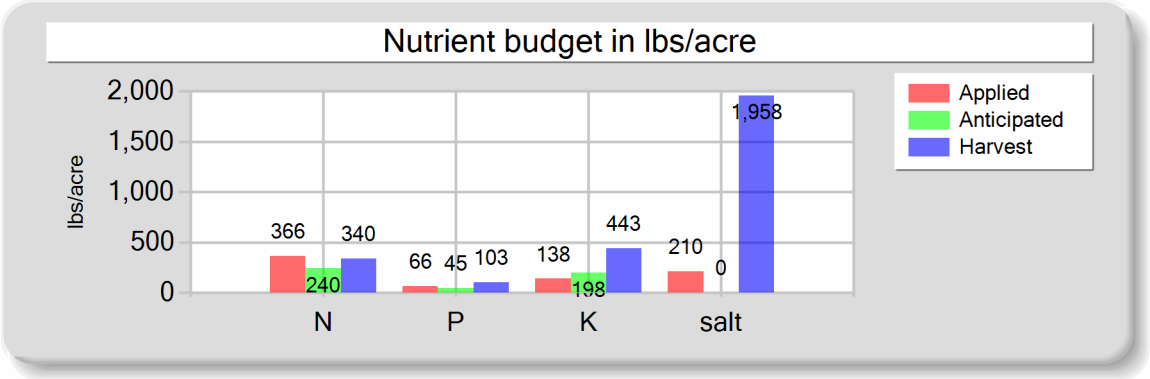
Fresh water applied	
6,447,640.00	*gallons*
237.44	*acre-inches*
11.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

53C - 05/16/2023: Corn, silage

Field name: 53C | Crop: Corn, silage | Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	358.96	65.66	137.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	210.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	365.96	65.66	137.89	210.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	339.95	103.34	443.29	1,958.08
Nutrient balance	26.02	-37.68	-305.39	-1,747.73
Applied to removed ratio	1.08	0.64	0.31	0.11

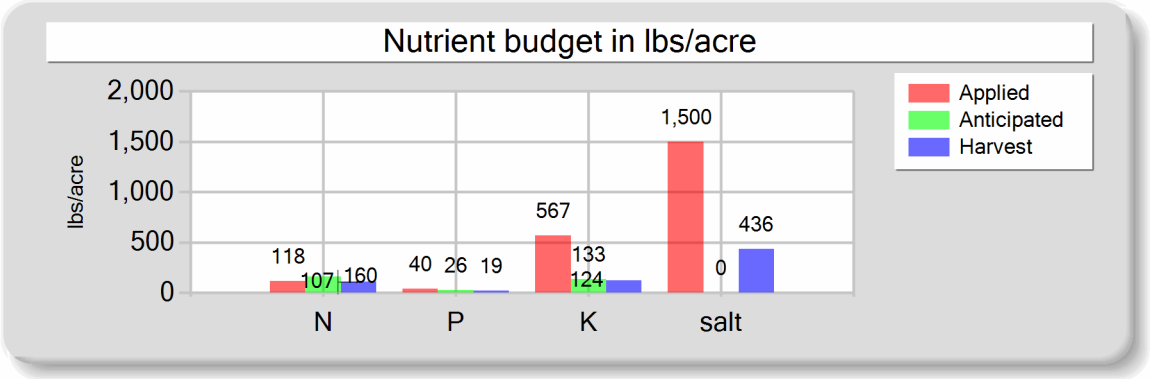
Fresh water applied	
14,403,920.00	*gallons*
530.45	*acre-inches*
26.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

54 - 11/17/2022: Oats, silage-soft dough

Field name: 54 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	113.10	39.71	566.91	1,406.04
Fresh water	0.00	0.00	0.00	94.30
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	117.76	39.71	566.91	1,500.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	107.03	19.29	124.45	435.57
Nutrient balance	10.74	20.42	442.46	1,064.76
Applied to removed ratio	1.10	2.06	4.56	3.44

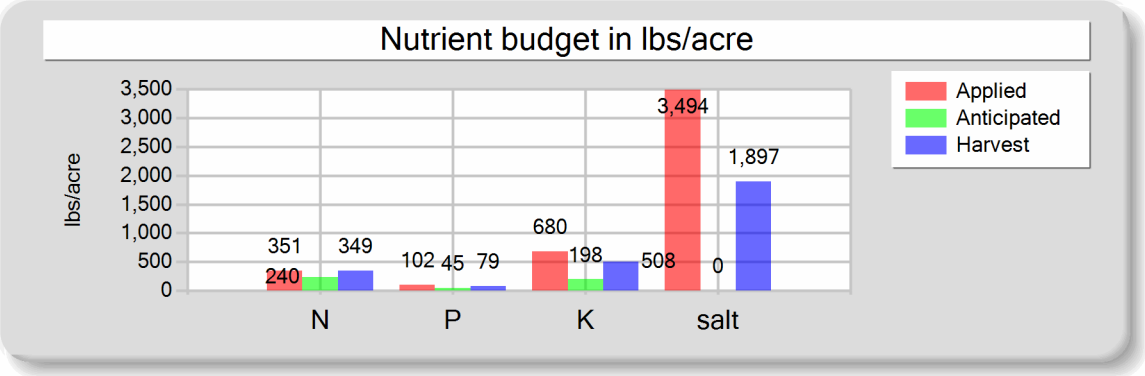
Fresh water applied	
13,883,080.00	*gallons*
511.27	*acre-inches*
11.89	*inches/acre*

Process wastewater applied	
5,796,000.00	*gallons*
213.45	*acre-inches*
4.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

54 - 05/16/2023: Corn, silage

Field name: 54 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	346.59	102.42	679.68	3,262.08
Fresh water	0.00	0.00	0.00	231.66
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	351.26	102.42	679.68	3,493.75
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	348.61	79.46	507.54	1,896.86
Nutrient balance	2.64	22.96	172.14	1,596.89
Applied to removed ratio	1.01	1.29	1.34	1.84

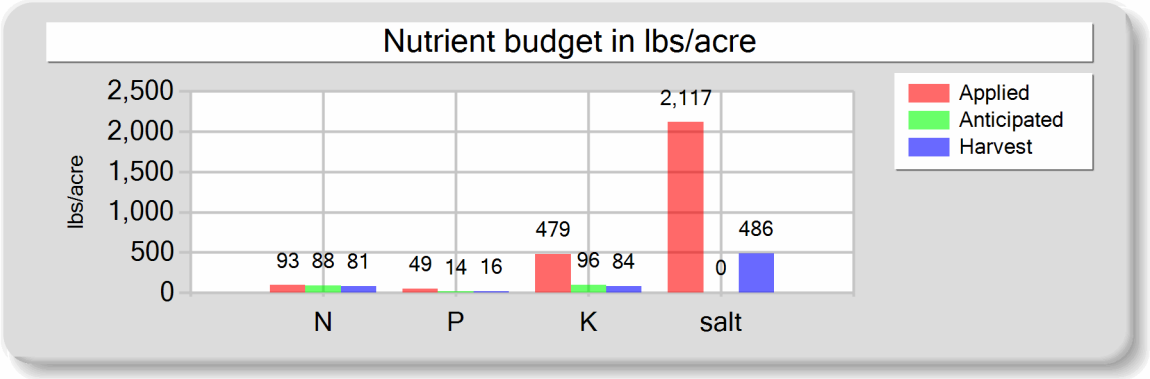
Fresh water applied	
34,106,040.00	*gallons*
1,256.01	*acre-inches*
29.21	*inches/acre*

Process wastewater applied	
8,454,000.00	*gallons*
311.33	*acre-inches*
7.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

54 - 09/01/2023: Sudangrass, silage

Field name: 54 Crop: Sudangrass, silage Plant date: 09/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	88.00	48.56	478.90	2,035.74
Fresh water	0.00	0.00	0.00	81.06
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	92.67	48.56	478.90	2,116.81
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	80.93	16.19	84.01	485.55
Nutrient balance	11.75	32.38	394.89	1,631.25
Applied to removed ratio	1.15	3.00	5.70	4.36

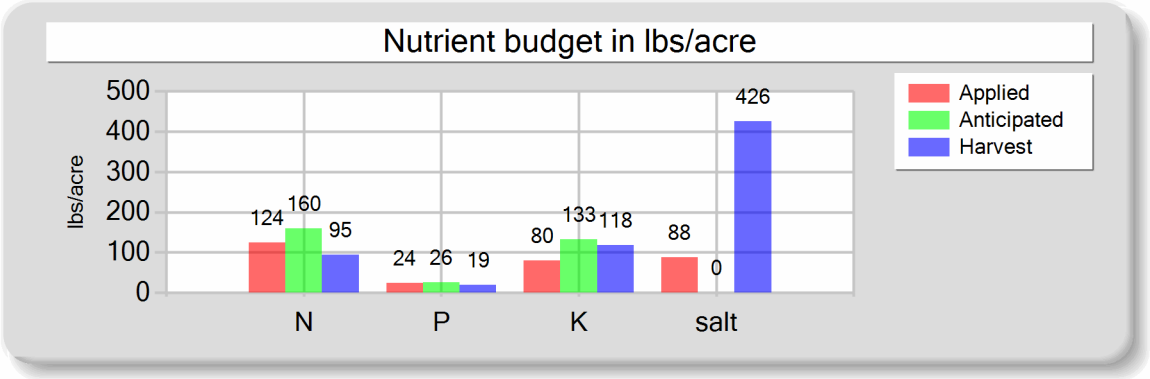
Fresh water applied	
11,934,420.00	*gallons*
439.50	*acre-inches*
10.22	*inches/acre*

Process wastewater applied	
5,430,000.00	*gallons*
199.97	*acre-inches*
4.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

55A - 11/16/2022: Oats, silage-soft dough

Field name: 55A | Crop: Oats, silage-soft dough | Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	117.41	23.62	79.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	88.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	124.41	23.62	79.89	88.49
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	94.54	19.26	117.88	426.01
Nutrient balance	29.87	4.36	-37.99	-337.53
Applied to removed ratio	1.32	1.23	0.68	0.21

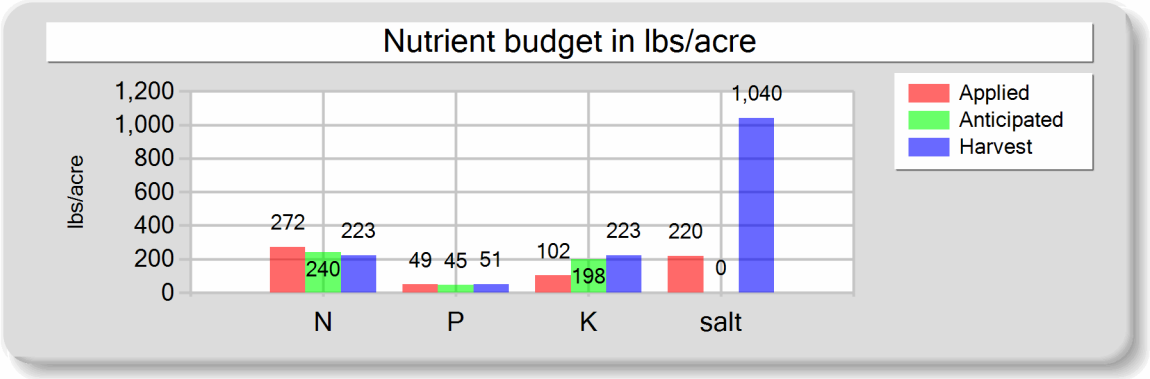
Fresh water applied	
5,756,180.00	*gallons*
211.98	*acre-inches*
11.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

55A - 05/16/2023: Corn, silage

Field name: 55A | Crop: Corn, silage | Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	265.16	48.51	101.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	219.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.16	48.51	101.86	219.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	223.05	51.24	223.05	1,039.90
Nutrient balance	49.11	-2.74	-121.19	-820.27
Applied to removed ratio	1.22	0.95	0.46	0.21

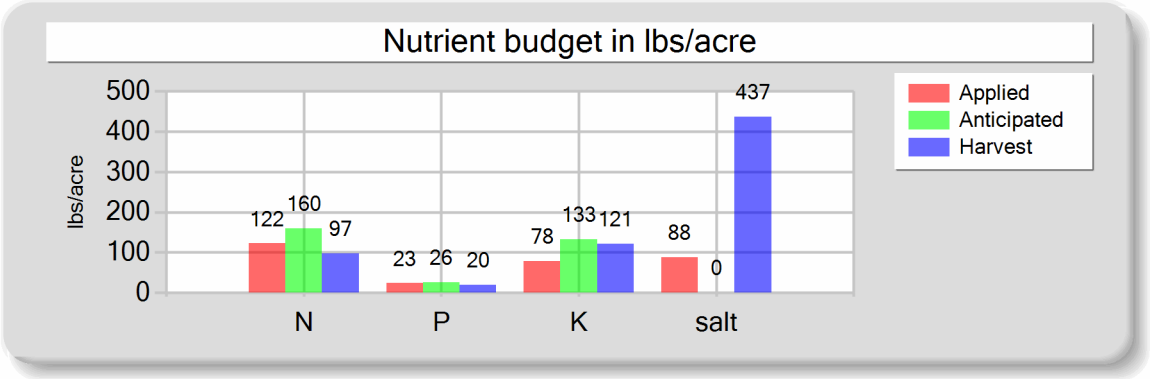
Fresh water applied	
14,287,180.00	*gallons*
526.15	*acre-inches*
27.69	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

55B - 11/16/2022: Oats, silage-soft dough

Field name: 55B | Crop: Oats, silage-soft dough | Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	115.26	23.19	78.43	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	88.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	122.26	23.19	78.43	88.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	96.92	19.74	120.85	436.75
Nutrient balance	25.34	3.44	-42.42	-348.76
Applied to removed ratio	1.26	1.17	0.65	0.20

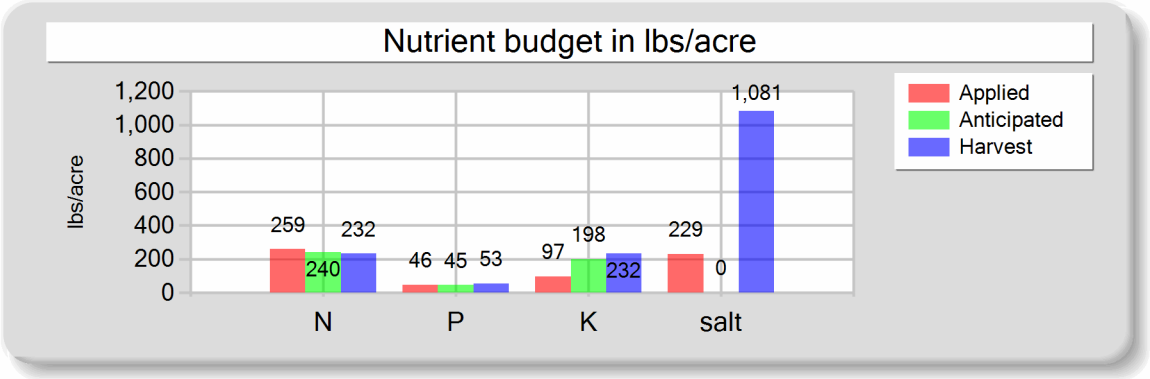
Fresh water applied	
6,025,580.00	*gallons*
221.90	*acre-inches*
11.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

55B - 05/16/2023: Corn, silage

Field name: 55B | Crop: Corn, silage | Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	251.90	46.08	96.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	228.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.90	46.08	96.77	228.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.97	53.29	231.97	1,081.50
Nutrient balance	26.93	-7.21	-135.21	-852.62
Applied to removed ratio	1.12	0.86	0.42	0.21

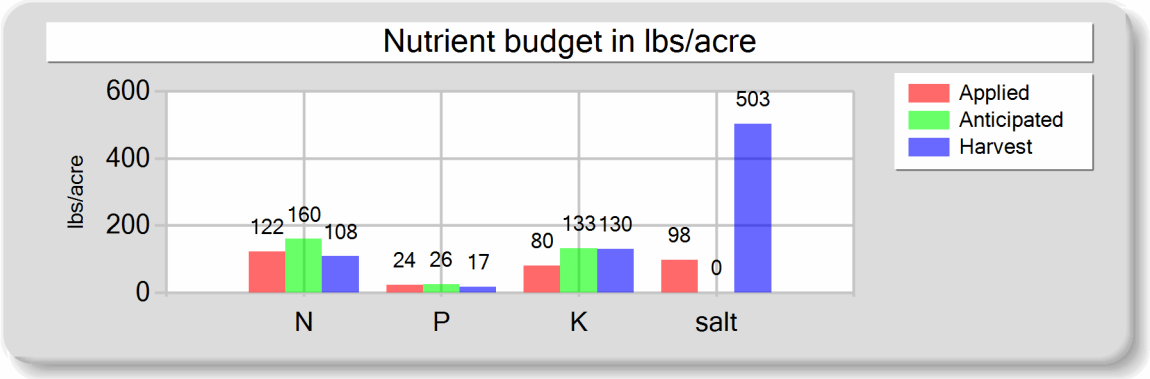
Fresh water applied	
14,888,840.00	*gallons*
548.31	*acre-inches*
28.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

56B - 11/16/2022: Oats, silage-soft dough

Field name: 56B Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	117.74	23.69	80.12	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	98.01
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	122.40	23.69	80.12	98.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	108.45	17.19	130.27	502.57
Nutrient balance	13.96	6.49	-50.15	-404.56
Applied to removed ratio	1.13	1.38	0.62	0.20

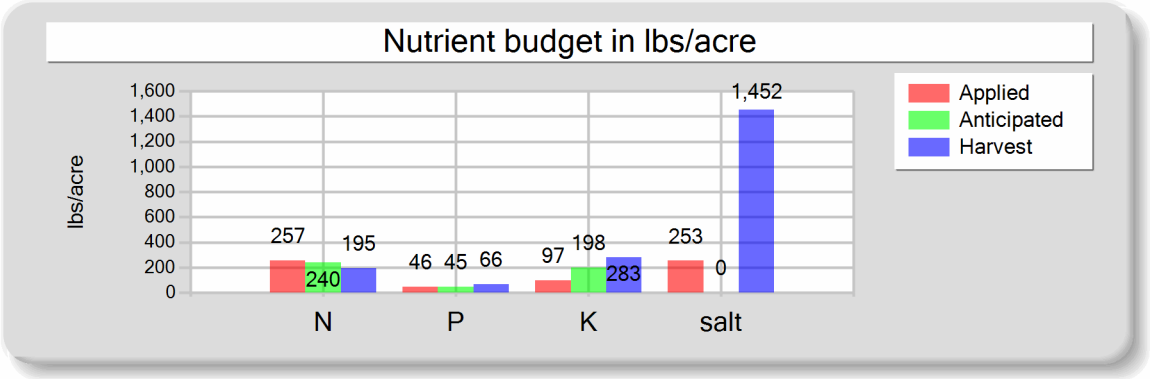
Fresh water applied	
20,133,160.00	*gallons*
741.44	*acre-inches*
12.36	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

56B - 05/15/2023: Corn, silage

Field name: 56B | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	251.90	46.08	96.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	253.06
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	256.57	46.08	96.77	253.06
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	195.36	66.31	283.18	1,451.72
Nutrient balance	61.22	-20.23	-186.41	-1,198.66
Applied to removed ratio	1.31	0.69	0.34	0.17

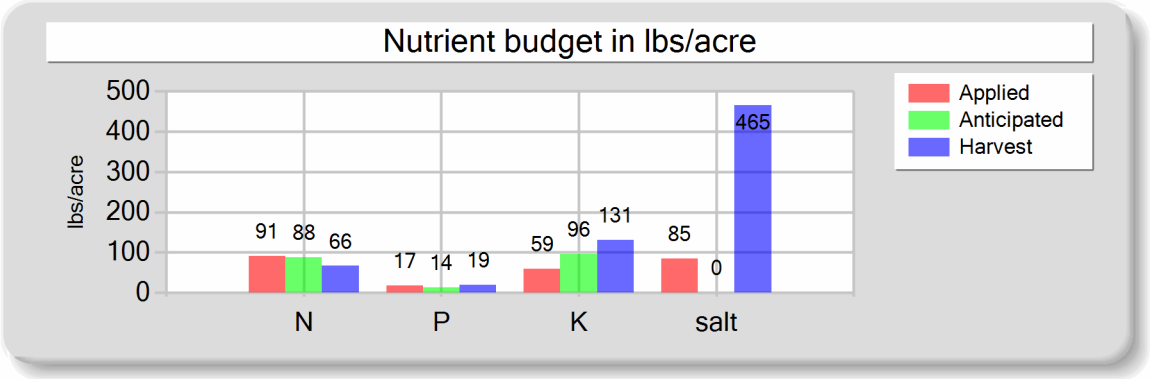
Fresh water applied	
51,985,220.00	*gallons*
1,914.44	*acre-inches*
31.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

56B - 08/25/2023: Sudangrass, silage

Field name: 56B | Crop: Sudangrass, silage | Plant date: 08/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	86.75	17.45	59.03	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	84.98
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	91.42	17.45	59.03	84.98
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	66.43	18.98	130.96	465.01
Nutrient balance	24.99	-1.53	-71.93	-380.03
Applied to removed ratio	1.38	0.92	0.45	0.18

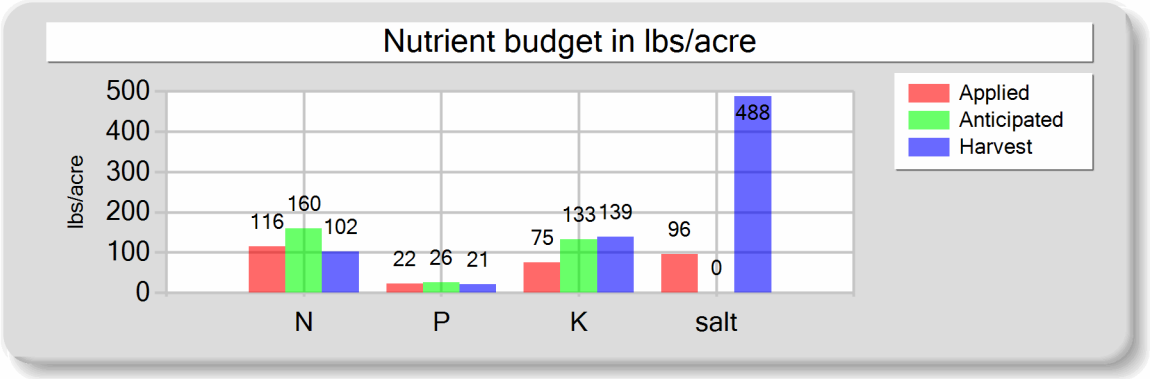
Fresh water applied	
17,457,120.00	*gallons*
642.89	*acre-inches*
10.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

57 - 11/17/2022: Oats, silage-soft dough

Field name: 57 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	110.89	22.31	75.46	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	95.53
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	115.55	22.31	75.46	95.53
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	102.02	21.44	138.99	487.94
Nutrient balance	13.53	0.87	-63.53	-392.41
Applied to removed ratio	1.13	1.04	0.54	0.20

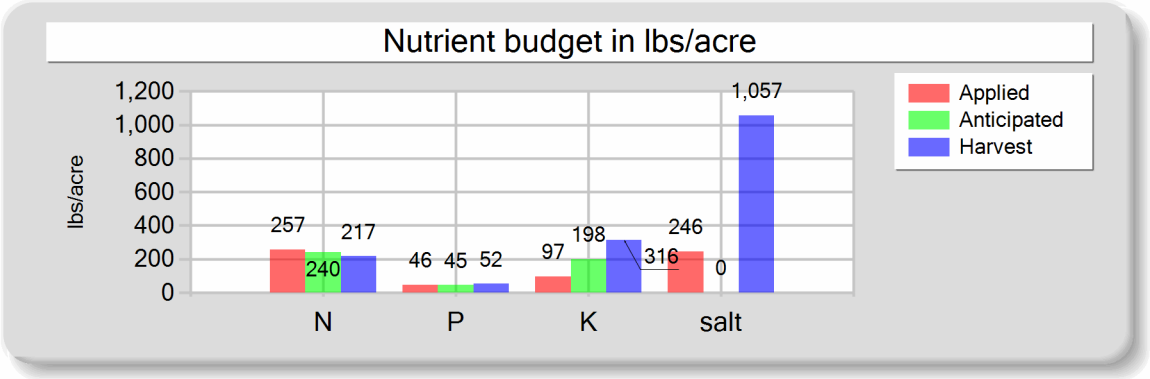
Fresh water applied	
18,642,480.00	*gallons*
686.54	*acre-inches*
12.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

57 - 05/15/2023: Corn, silage

Field name: 57 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	251.90	46.08	96.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	245.86
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	256.57	46.08	96.77	245.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	216.51	52.03	315.53	1,057.35
Nutrient balance	40.07	-5.95	-218.76	-811.49
Applied to removed ratio	1.19	0.89	0.31	0.23

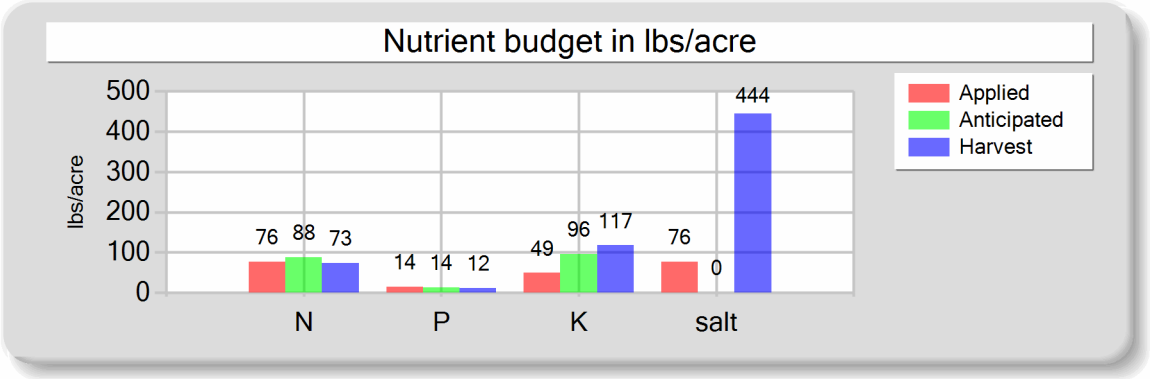
Fresh water applied	
47,980,140.00	*gallons*
1,766.95	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

57 - 09/01/2023: Sudangrass, silage

Field name: 57 Crop: Sudangrass, silage Plant date: 09/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	71.75	14.44	48.82	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	76.38
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	76.42	14.44	48.82	76.38
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	73.29	11.91	117.34	444.09
Nutrient balance	3.12	2.52	-68.52	-367.71
Applied to removed ratio	1.04	1.21	0.42	0.17

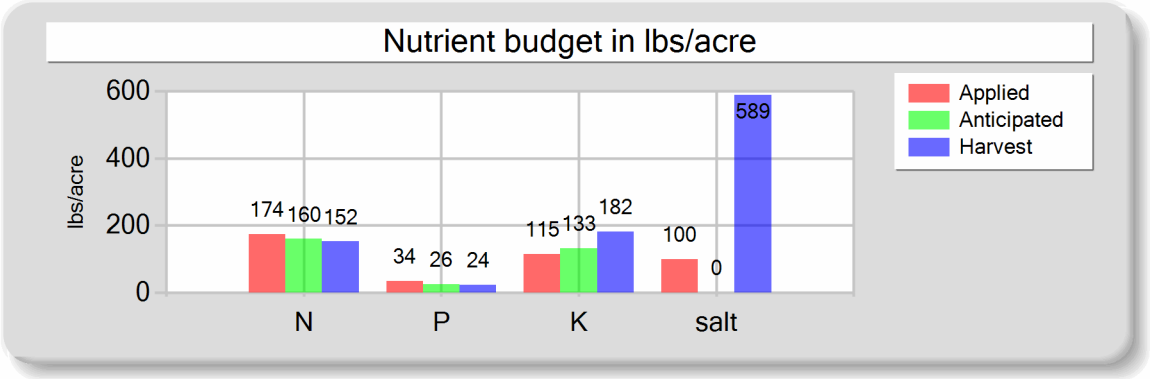
Fresh water applied	
14,906,800.00	*gallons*
548.97	*acre-inches*
9.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

59A - 11/18/2022: Oats, silage-soft dough

Field name: 59A Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.00	34.00	115.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	100.38
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	173.67	34.00	115.00	100.38
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.45	23.57	181.91	589.20
Nutrient balance	21.21	10.43	-66.91	-488.81
Applied to removed ratio	1.14	1.44	0.63	0.17

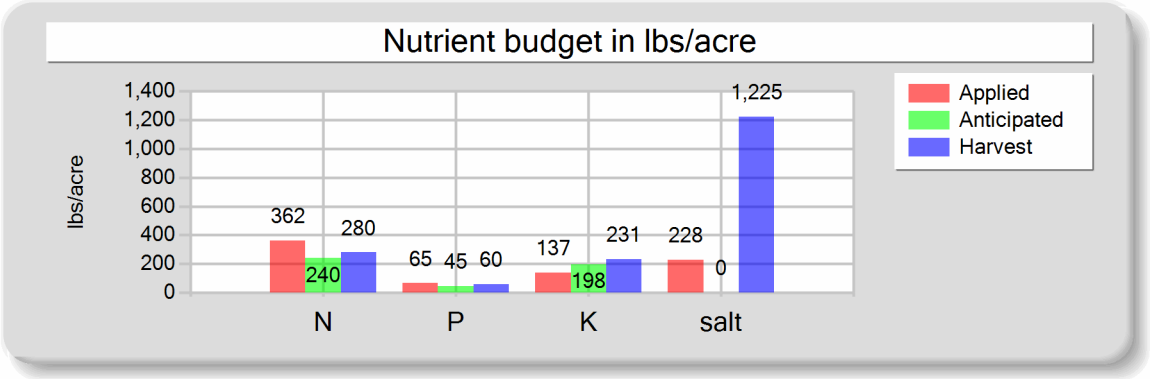
Fresh water applied	
15,122,320.00	*gallons*
556.90	*acre-inches*
12.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

59A - 05/15/2023: Corn, silage

Field name: 59A | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	357.82	65.45	137.45	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	227.83
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	362.48	65.45	137.45	227.83
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	279.89	60.25	231.30	1,224.54
Nutrient balance	82.59	5.20	-93.85	-996.71
Applied to removed ratio	1.30	1.09	0.59	0.19

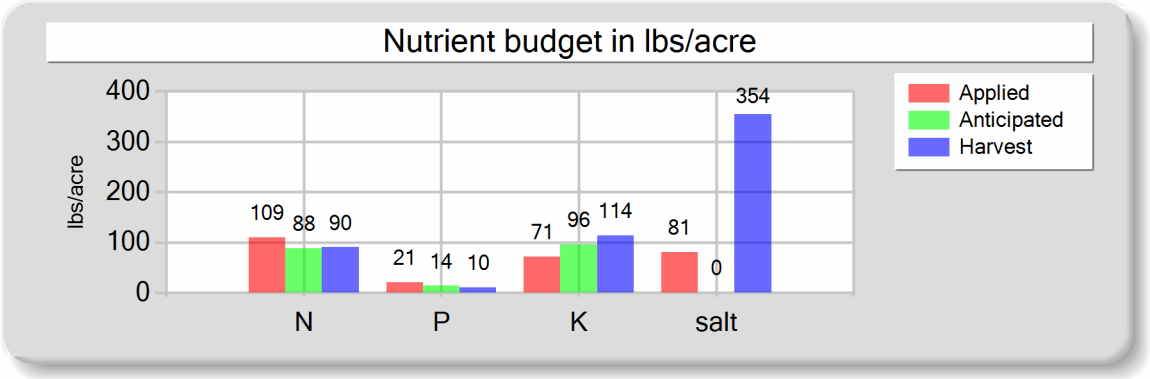
Fresh water applied	
34,321,560.00	*gallons*
1,263.95	*acre-inches*
28.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

59A - 08/23/2023: Sudangrass, silage

Field name: 59A | Crop: Sudangrass, silage | Plant date: 08/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	104.78	21.08	71.30	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	81.01
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	109.45	21.08	71.30	81.01
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	90.18	10.17	114.12	354.16
Nutrient balance	19.27	10.91	-42.82	-273.15
Applied to removed ratio	1.21	2.07	0.62	0.23

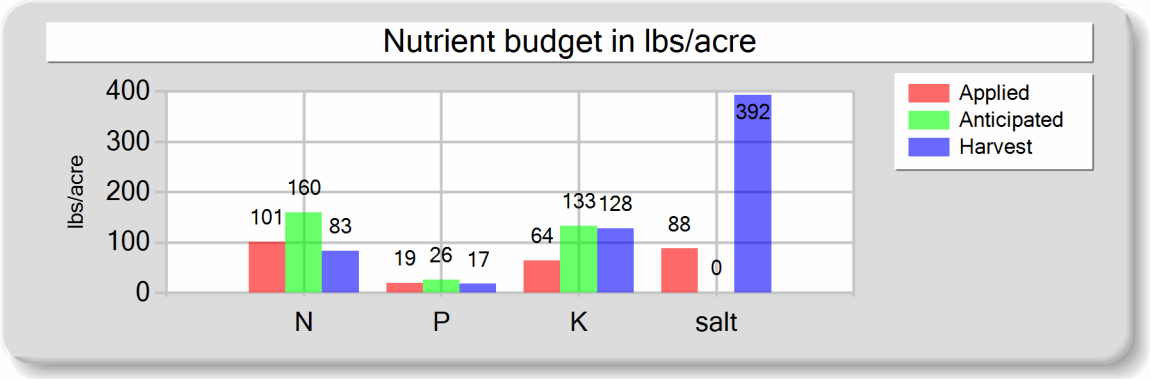
Fresh water applied	
12,203,820.00	*gallons*
449.43	*acre-inches*
10.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

59B - 11/18/2022: Oats, silage-soft dough

Field name: 59B | Crop: Oats, silage-soft dough | Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	93.84	18.88	63.85	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	87.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	100.84	18.88	63.85	87.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	83.40	17.43	127.58	392.09
Nutrient balance	17.44	1.45	-63.73	-304.54
Applied to removed ratio	1.21	1.08	0.50	0.22

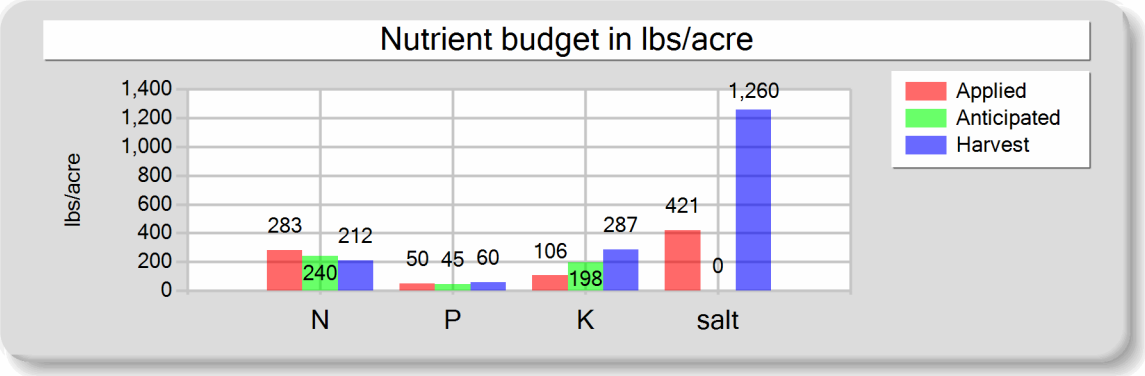
Fresh water applied	
12,589,960.00	*gallons*
463.65	*acre-inches*
11.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

59B - 06/15/2023: Corn, silage

Field name: 59B | Crop: Corn, silage | Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	275.89	50.47	105.98	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	420.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.89	50.47	105.98	420.77
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	211.77	59.51	287.03	1,260.11
Nutrient balance	71.13	-9.04	-181.04	-839.34
Applied to removed ratio	1.34	0.85	0.37	0.33

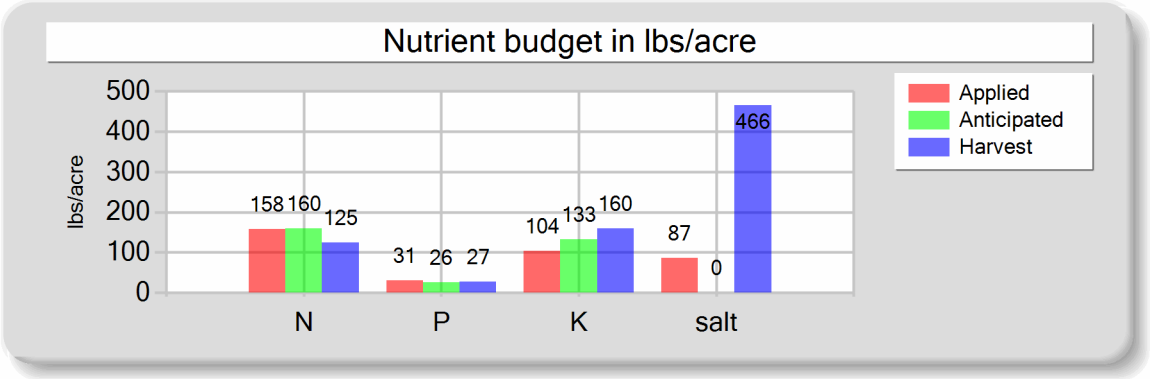
Fresh water applied	
33,315,800.00	*gallons*
1,226.91	*acre-inches*
29.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

60 - 11/18/2022: Oats, silage-soft dough

Field name: 60 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	153.09	30.80	104.18	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	86.71
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	157.76	30.80	104.18	86.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	124.81	26.87	160.04	465.78
Nutrient balance	32.95	3.93	-55.86	-379.08
Applied to removed ratio	1.26	1.15	0.65	0.19

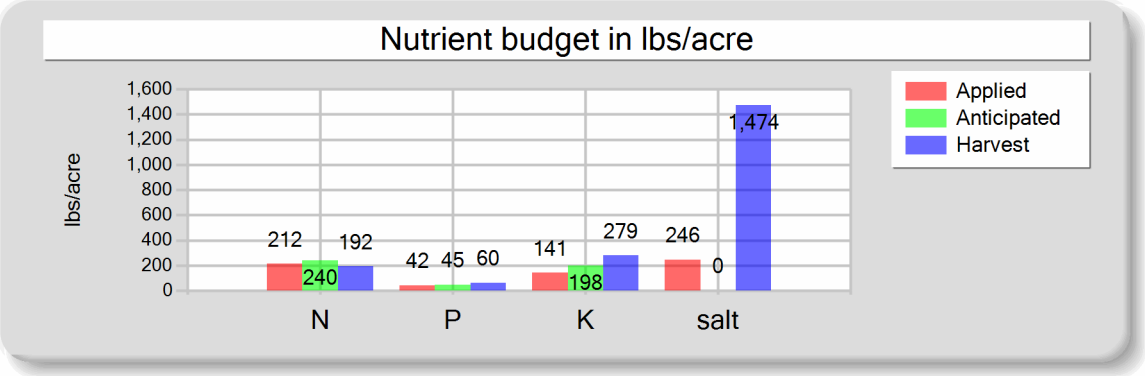
Fresh water applied	
10,093,520.00	*gallons*
371.71	*acre-inches*
10.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

60 - 05/15/2023: Corn, silage

Field name: 60 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	207.77	41.80	141.38	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	245.62
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	212.44	41.80	141.38	245.62
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	191.98	60.12	279.25	1,473.81
Nutrient balance	20.45	-18.32	-137.86	-1,228.18
Applied to removed ratio	1.11	0.70	0.51	0.17

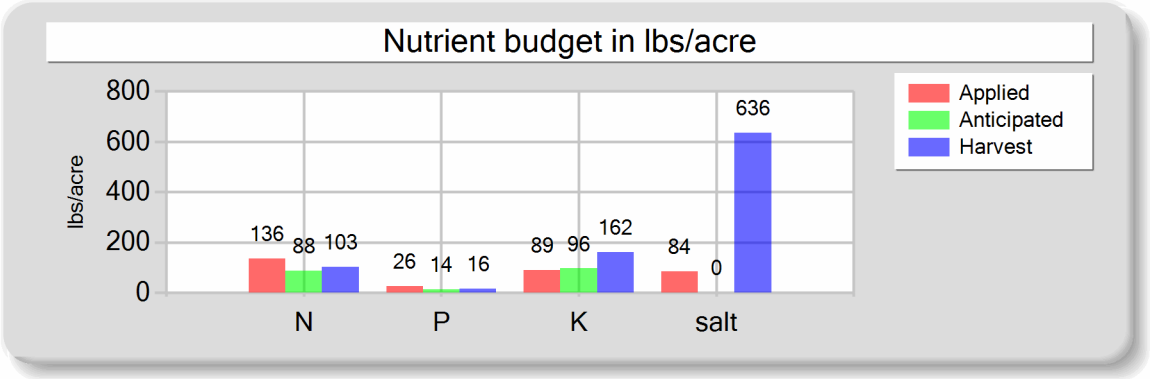
Fresh water applied	
28,592,320.00	*gallons*
1,052.96	*acre-inches*
30.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

60 - 08/24/2023: Sudangrass, silage

Field name: 60 Crop: Sudangrass, silage Plant date: 08/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	131.22	26.40	89.29	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	84.08
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	135.89	26.40	89.29	84.08
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	103.04	16.49	161.93	635.93
Nutrient balance	32.85	9.91	-72.63	-551.84
Applied to removed ratio	1.32	1.60	0.55	0.13

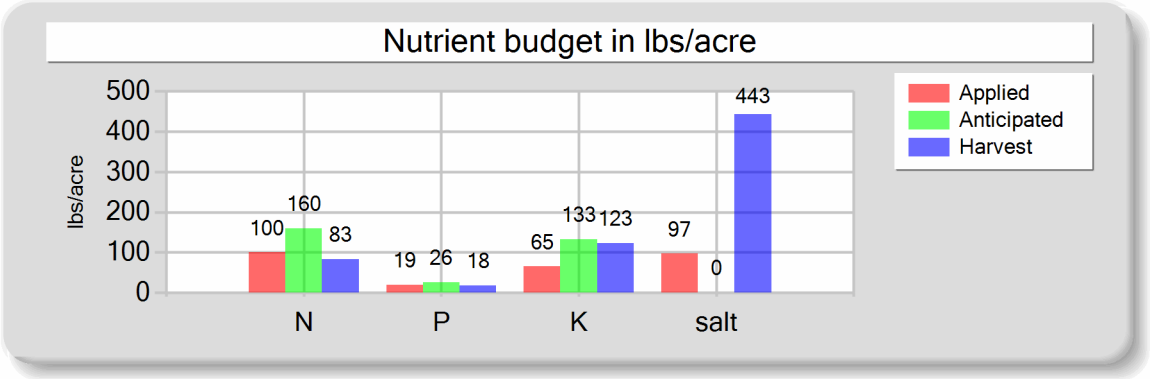
Fresh water applied	
9,788,200.00	*gallons*
360.47	*acre-inches*
10.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Richards - 11/18/2022: Oats, silage-soft dough

Field name: Richards | Crop: Oats, silage-soft dough | Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	95.33	19.18	64.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	96.74
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	100.00	19.18	64.87	96.74
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	83.43	17.72	123.30	442.98
Nutrient balance	16.57	1.46	-58.42	-346.24
Applied to removed ratio	1.20	1.08	0.53	0.22

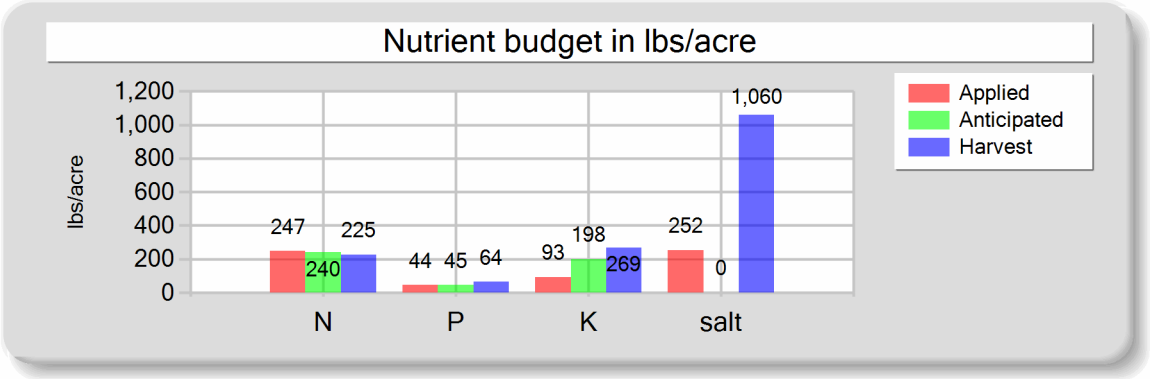
Fresh water applied	
25,835,460.00	*gallons*
951.43	*acre-inches*
12.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Richards - 05/16/2023: Corn, silage

Field name: Richards | Crop: Corn, silage | Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.22	44.31	93.05	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	251.59
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	246.88	44.31	93.05	251.59
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.95	63.57	268.96	1,059.54
Nutrient balance	21.93	-19.26	-175.91	-807.95
Applied to removed ratio	1.10	0.70	0.35	0.24

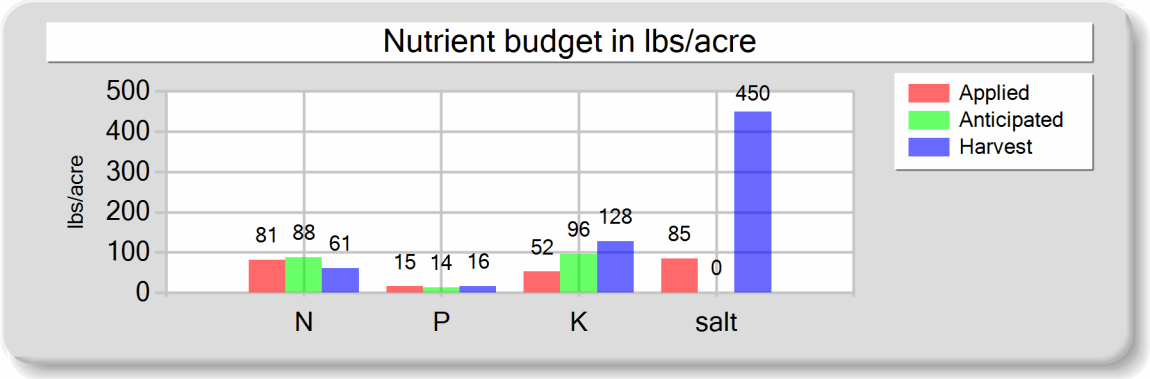
Fresh water applied	
67,188,360.00	*gallons*
2,474.32	*acre-inches*
31.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Richards - 08/25/2023: Sudangrass, silage

Field name: Richards | Crop: Sudangrass, silage | Plant date: 08/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	76.27	15.34	51.90	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	84.87
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	80.93	15.34	51.90	84.87
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	60.77	15.80	127.61	449.66
Nutrient balance	20.17	-0.46	-75.71	-364.79
Applied to removed ratio	1.33	0.97	0.41	0.19

Fresh water applied	
22,665,520.00	*gallons*
834.69	*acre-inches*
10.70	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid manure

Sample and source description: Solid manure

Sample date: 06/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,000.00	6,300.00							0.00
DL	0.01	0.01	0.01							0.01

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	3,400.00	11,500.00							0.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	245.00	142.00	0.00	1.10	57.90	388.00								3,710.00	1,830
DL	1.00	0.50	0.01	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	33.70	0.00	0.00	0.80	21.70	439.00								2,470.00	1,560
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	134.00	67.10	0.00	2.00	72.20	471.00								3,120.00	2,330
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	100.00	63.00	0.00	0.00	35.30	504.00								3,660.00	1,250
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - 60-WD

Domestic Well - 60-WD

60-WD

Sample description: 60-WD

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										346.00	
DL	0.10										10.00	

Irrigation Water - Surface Water

Surface Water

Sample description: Surface Water

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										27.40	35
DL	0.10										10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

53B - 11/17/2022: Oats, silage-soft dough

53B - 11/17/2022: Oats, silage-soft dough

53B & 53C

Sample and source description: 53B & 53C

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	3,000.00	19,700.00		6.20
DL	100.00	100.00	100.00		0.01

53B - 05/16/2023: Corn, silage

53B & 53C

Sample and source description: 53B & 53C

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	3,800.00	16,300.00		7.20
DL	100.00	100.00	100.00		0.01

53C - 11/17/2022: Oats, silage-soft dough

53B & 53C

Sample and source description: 53B & 53C

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	3,000.00	19,700.00		6.20
DL	100.00	100.00	100.00		0.01

53C - 05/16/2023: Corn, silage

53B & 53C

Sample and source description: 53B & 53C

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	3,800.00	16,300.00		7.20
DL	100.00	100.00	100.00		0.01

54 - 11/17/2022: Oats, silage-soft dough

53A + 54

Sample and source description: 53A + 54

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,100.00	20,000.00		7.00
DL	100.00	100.00	100.00		0.01

54 - 05/16/2023: Corn, silage

54

Sample and source description: 54

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	3,100.00	19,800.00		7.40
DL	100.00	100.00	100.00		0.01

54 - 09/01/2023: Sudangrass, silage

54

Sample and source description: 54

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	4,200.00	21,800.00		12.60
DL	100.00	100.00	100.00		0.01

55A - 11/16/2022: Oats, silage-soft dough

55A + 55B

Sample and source description: 55A + 55B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,300.00	20,200.00		7.30
DL	100.00	100.00	100.00		0.01

55A - 05/16/2023: Corn, silage

55A & 55B

Sample and source description: 55A & 55B

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	3,400.00	14,800.00		6.90
DL	100.00	100.00	100.00		0.01

55B - 11/16/2022: Oats, silage-soft dough

55A + 55B

Sample and source description: 55A + 55B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,300.00	20,200.00		7.30
DL	100.00	100.00	100.00		0.01

55B - 05/16/2023: Corn, silage

55A & 55B

Sample and source description: 55A & 55B

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	3,400.00	14,800.00		6.90
DL	100.00	100.00	100.00		0.01

56B - 11/16/2022: Oats, silage-soft dough

56B

Sample and source description: 56B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	2,600.00	19,700.00		7.60
DL	100.00	100.00	100.00		0.01

56B - 05/15/2023: Corn, silage

56B

Sample and source description: 56B

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	3,700.00	15,800.00		8.10
DL	100.00	100.00	100.00		0.01

56B - 08/25/2023: Sudangrass, silage

56B

Sample and source description: 56B

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	4,000.00	27,600.00		9.80
DL	100.00	100.00	100.00		0.01

57 - 11/17/2022: Oats, silage-soft dough

57A

Sample and source description: 57A

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	3,100.00	18,700.00		6.80
DL	100.00	100.00	100.00		0.01

57 - 11/17/2022: Oats, silage-soft dough

57B

Sample and source description: 57B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,900.00	18,800.00		6.60
DL	100.00	100.00	100.00		0.01

57 - 05/15/2023: Corn, silage

57A

Sample and source description: 57A

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	3,100.00	18,800.00		6.30
DL	100.00	100.00	100.00		0.01

57B

Sample and source description: 57B

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	3,500.00	18,000.00		6.30
DL	100.00	100.00	100.00		0.01

57 - 09/01/2023: Sudangrass, silage

57 - 09/01/2023: Sudangrass, silage

57A

Sample and source description: 57A

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	3,300.00	32,500.00		12.30
DL	100.00	100.00	100.00		0.01

57B

Sample and source description: 57B

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,600.00	3,300.00	36,500.00		12.00
DL	100.00	100.00	100.00		0.01

59A - 11/18/2022: Oats, silage-soft dough

59A

Sample and source description: 59A

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	3,200.00	24,700.00		8.00
DL	100.00	100.00	100.00		0.01

59A - 05/15/2023: Corn, silage

59A - 05/15/2023: Corn, silage

59A

Sample and source description: 59A

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	3,100.00	11,900.00		6.30
DL	100.00	100.00	100.00		0.01

59A - 08/23/2023: Sudangrass, silage

59A

Sample and source description: 59A

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	3,100.00	34,800.00		10.80
DL	100.00	100.00	100.00		0.01

59B - 11/18/2022: Oats, silage-soft dough

59B

Sample and source description: 59B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,800.00	20,500.00		6.30
DL	100.00	100.00	100.00		0.01

59B - 06/15/2023: Corn, silage

59B

Sample and source description: 59B

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,400.00	16,400.00		7.20
DL	100.00	100.00	100.00		0.01

60 - 11/18/2022: Oats, silage-soft dough

60

Sample and source description: 60

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	4,500.00	26,800.00		7.80
DL	100.00	100.00	100.00		0.01

60 - 05/15/2023: Corn, silage

60

Sample and source description: 60

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	3,100.00	14,400.00		7.60
DL	100.00	100.00	100.00		0.01

60 - 08/24/2023: Sudangrass, silage

60

Sample and source description: 60

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	2,800.00	27,500.00		10.80
DL	100.00	100.00	100.00		0.01

Richards - 11/18/2022: Oats, silage-soft dough

Richards - A

Sample and source description: Richards - A

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,400.00	16,700.00		6.00
DL	100.00	100.00	100.00		0.01

Richards - B

Sample and source description: Richards - B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,500.00	16,800.00		5.90
DL	100.00	100.00	100.00		0.01

Richards - 05/16/2023: Corn, silage

Richards - 05/16/2023: Corn, silage

Richards - A

Sample and source description: Richards - A

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	4,100.00	17,200.00		8.20
DL	100.00	100.00	100.00		0.01

Richards - B

Sample and source description: Richards - B

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	3,900.00	16,500.00		6.50
DL	100.00	100.00	100.00		0.01

Richards - 08/25/2023: Sudangrass, silage

Richards - A

Sample and source description: Richards - A

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,700.00	29,200.00		11.60
DL	100.00	100.00	100.00		0.01

Richards - 08/25/2023: Sudangrass, silage

Richards - B

Sample and source description: Richards - B

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	3,900.00	31,500.00		11.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

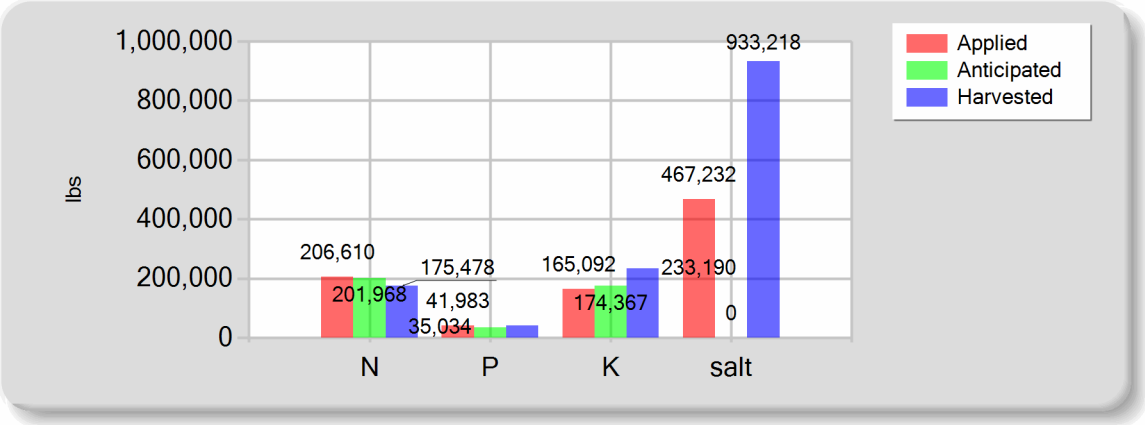
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

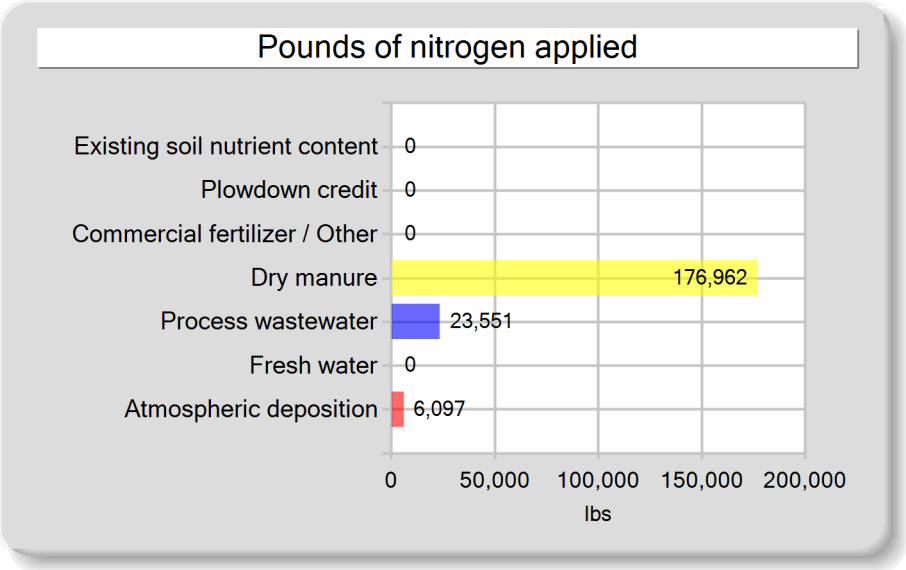
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

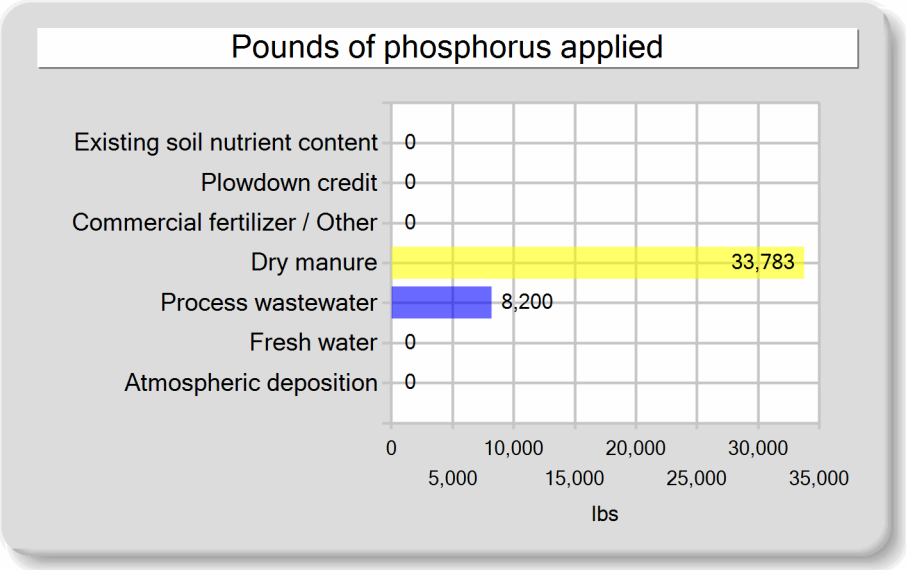
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176,962.43	33,783.04	90,895.74	0.00
Process wastewater	23,550.74	8,199.75	74,196.38	288,266.01
Fresh water	0.00	0.00	0.00	178,966.15
Atmospheric deposition	6,097.00	0.00	0.00	0.00
Total nutrients applied	206,610.17	41,982.79	165,092.12	467,232.16
Anticipated crop nutrient removal	201,968.00	35,034.20	174,366.80	0.00
Actual crop nutrient removal	175,477.89	41,934.40	233,190.28	933,217.68
Nutrient balance	31,132.28	48.39	-68,098.15	-465,985.52
Applied to removed ratio	1.18	1.00	0.71	0.50

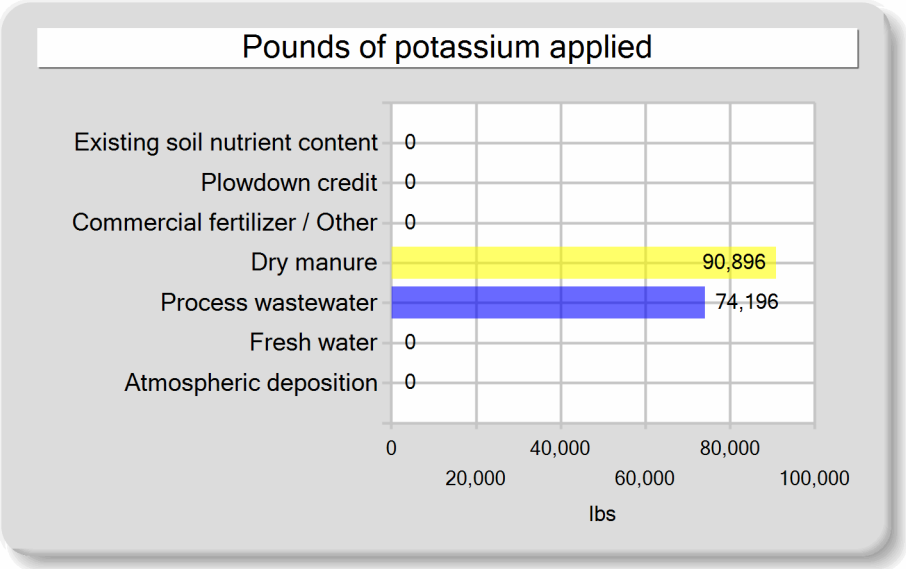
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

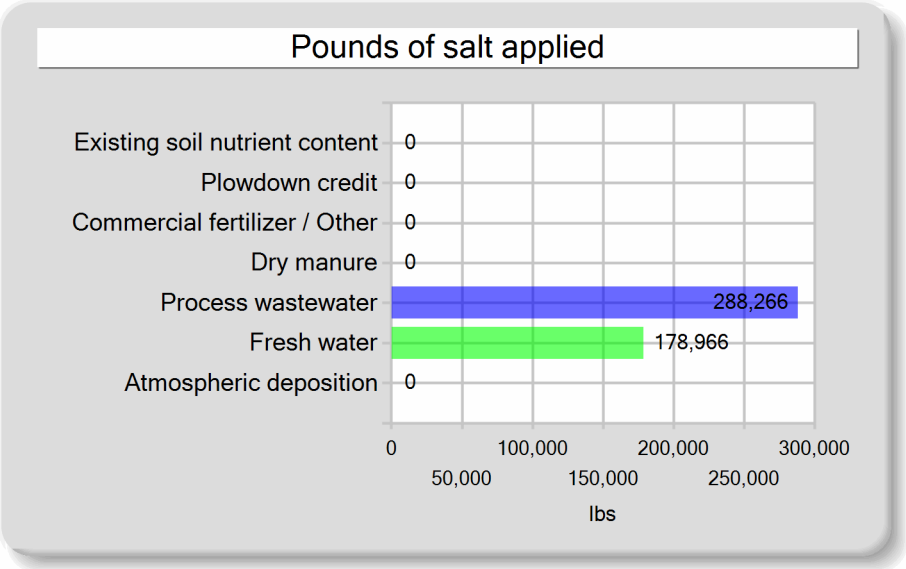


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes
Lloyd Fagundes (Jun 25, 2024 17:06 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Fagundes Bros. Dairy C/O Ralph, Lloyd, &/or Fred Fagundes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/25/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #2

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:52

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0558-01	60-WD	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: 60-WD
23H0558-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.35**	mmhos/cm	0.01	1		08/04/23 14:36	SM 2510 B		BEH0246
Electrical Conductivity umhos	**346**	umhos/cm	10.0	1		08/04/23 14:36	SM 2510 B		BEH0246
Ammonia (as N)	ND	mg/L	0.00	1		08/02/23 16:00	Field		BEH0306
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/03/23 19:30	EPA 300.0		BEH0186
pH	**7.5**	units	1.0	1		08/04/23 14:36	SM 4500-H+	H	BEH0246
Temperature	**25.0**	°C	0.0	1		08/04/23 14:36	SM 2510 B		BEH0246

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0186										
Blank (BEH0186-BLK1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0186-BLK2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0186-BS1)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		93.5	90-110		
Duplicate (BEH0186-DUP1)		**Source: 23H0558-01**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	0.07		0.1	mg/L		0.07			1.48	10
Matrix Spike (BEH0186-MS1)		**Source: 23H0558-01**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.07	92.2	90-110		
Reference (BEH0186-SRM1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.7	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0246										
Blank (BEH0246-BLK1)					Prepared & Analyzed: 8/4/2023					
pH	5.5		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0246-BLK2)					Prepared & Analyzed: 8/4/2023					
pH	7.3		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0246-BLK3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0246-DUP1)		**Source: 23H0547-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.03		0.01	mmhos/cm		0.03			8.26	10
pH	7.8		1.0	units		7.8			0.383	10
Electrical Conductivity umhos	26.7		10.0	umhos/cm		29.0			8.26	10
Duplicate (BEH0246-DUP2)		**Source: 23H0561-03**			Prepared & Analyzed: 8/4/2023					
pH	8.0		1.0	units		8.0			0.251	10
Electrical Conductivity	0.33		0.01	mmhos/cm		0.32			0.277	10
Electrical Conductivity umhos	325		10.0	umhos/cm		324			0.277	10
Reference (BEH0246-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	547			umhos/cm	538.0		102	90-110		
Reference (BEH0246-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.8			units	7.790		101	.7163-101.28		
Reference (BEH0246-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0246-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0246-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0246 (Continued)										
Reference (BEH0246-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0246-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0246-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0246-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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JG

08/03/23 15:10 23H0558

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 1 No. of Bottles: 1

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Fagundes Brothers Dairy #2
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) **DWW1: EC, pH, NO_3-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	60-WD	8/2/2023	1600	-1.2	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JLF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0558

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice:☒ Blue Ice:☐ None:☐

Container(s) Received	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										